ALFFRIN REGI

Kerala, India

+91 9594937829 alffrin10@gmail.com linkedin.com/in/alffrin-regi

Education

Lovely Professional University
Bachelors of Technology in Robotics and Automation

Technical Skills

- ROS 2
- C++, Python
- Power Electronics
- Signal Processing
- Sensor Fusion
- PCB Designing
- Embedded Systems
- Vision Systems

Soft Skills

- Creative
- Adaptable
- Resilience
- Quick-Learner

Experience

Zaalima Development — **Mar 2024 – Jun 2024**
Robotics and Embedded System Intern — Bangalore, India

Renovation Tech Robotics Aeromodelling — **Sep 2023 – Nov 2025**
Technical Head — Punjab, India

Robotics and Intelligent System Community — **Sep 2023 – Nov 2025**
Technical Member — Punjab, India

Projects

Disaster Response Drone | Jetson Nano, YOLO

- Developed an autonomous drone capable of human detection using Jetson Nano and optimized YOLO model
- Integrated onboard vision processing and sensor fusion for stability and robust field deployment
- Designed the platform for disaster scenario with a focus on endurance, efficient hardware and enhanced decision-making algorithm

Landmine Detection Rover | RF Frequency, MATLAB, GPRPy

- Developed a ground robot equipped with custom GPR sensing pipeline for subsurface anomaly detection
- Designed the full RF chain including LNA, couplers, attenuators, and signal conditioning hardware for mid-frequency Ground Penetrating Radar (GPR)
- Built a processing workflow using MATLAB and GPRPy to visualize underground signatures and supporting real-time interpretation

Cybersecurity LLM with whisper Interface | LLM Fine-tuning, NLP, Python

- Fine-tuned a lightweight LLM specializing in cybersecurity threat detection & analysis, and automated response workflows
- Integrated Whisper for speech-based interaction, enabling voice-driven log queries
- Improved model for low-latency interference and secure on -device processing of sensitive data

Vision-Enabled Quadruped Robot | Computer Vision, Raspberry Pi

- Built a quadruped robot capable of perception-based navigation using onboard computer vision and real-time interference
- Implemented multi-sensor fusion for environmental understanding and terrain adaptation